

Non-Parametric Insurance Loss Modelling using Variable-Knot Splines

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Data

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DDFS (Density & Distribution Func. variable-knot Spline estimation)

Novel non-parametric method for *simultaneous* variable knot spline estimation of both the pdf and the cdf of a random variable.

- ① In the literature, non-parametric density estimation methods address the estimation of *solely the pdf*. However, pdf and cdf are closely connected.
- ② Model structure under which, both the pdf and cdf spline models share the *same set of knots* + their *coefficients are connected*.
- ③ Iterative approach combining:
 - **Constrained maximum likelihood estimation** of the spline coefficients.
 - **Sequential minimum bias driven estimation** of the underlying spline knots, following the Geometrically Designed Spline (GeDS) methodology (Kaishev et al., 2016, Dimitrova et al., 2023, Dimitrova et al., 2025).
- ④ Competitive alternative to state of the art methods, e.g., kernel, logsplines (Kooperberg and Stone, 1991), and recent spline-based methods (Cui et al., 2020, Kirkby et al., 2021) + *Large sample properties*.

Iterative estimation process

Step 1. Given $\mathbf{t}_{k,n}$, find the MLE estimates, $\hat{\boldsymbol{\theta}}$, of $f(x; \mathbf{t}_{k,n}, \boldsymbol{\theta})$.

Step 2. Use $\hat{\boldsymbol{\theta}}$ and $\mathbf{t}_{k,n}$ to compute $\hat{\boldsymbol{\theta}}'$, and then compute $F(x; \mathbf{t}_{k,n+1}, \hat{\boldsymbol{\theta}}')$, recalling that, $\mathbf{t}_{k,n+1}$ is obtained from $\mathbf{t}_{k,n}$, by adding the additional end knots, $t_0 = t_1$, and $t_{2n+k+1} = t_{2n+k}$.

Step 3. Compute the residuals

$$\rho_i = F_N(X_i) - F(X_i; \mathbf{t}_{k,n+1}, \hat{\boldsymbol{\theta}}'), \quad i = 1, \dots, N, \quad (1)$$

that provide information for the discrepancy between the ecdf, $F_N(x)$, and the estimate, $F(X_i; \mathbf{t}_{k,n+1}, \hat{\boldsymbol{\theta}}')$, of the cdf F .

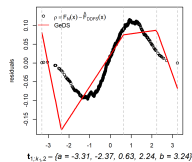
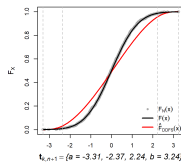
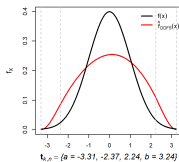
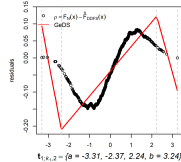
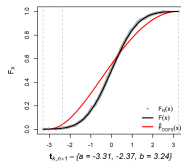
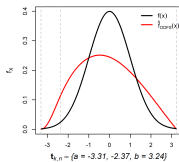
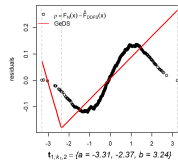
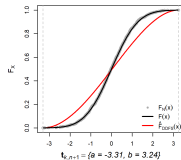
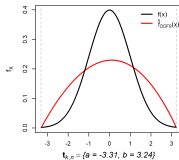
On the k -th iteration, if $k \leq q$, go to Step 4. Otherwise, use the residuals, ρ_i , $i = 1, \dots, N$, to compute the ratio

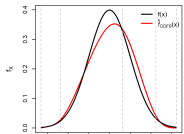
$$\phi = \frac{\sum_{i=1}^N \left\{ F_N(X_i) - F(X_i; \mathbf{t}_{k,n+1}, \hat{\boldsymbol{\theta}}') \right\}^2}{\sum_{i=1}^N \left\{ F_N(X_i) - F(X_i; \mathbf{t}_{k-q,n+1}, \hat{\boldsymbol{\theta}}') \right\}^2}. \quad (2)$$

If $\phi \geq \phi_{exit}$, then exit the iterations with final estimates, $f(x; \mathbf{t}_{k-q,n}, \hat{\boldsymbol{\theta}})$ and $F(x; \mathbf{t}_{k-q,n+1}, \hat{\boldsymbol{\theta}}')$. If $\phi < \phi_{exit}$, then go to Step 4.

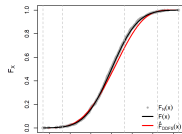
Step 4. Find a new knot δ^* , applying the *locally adaptive bias minimizing knot insertion scheme* of Stage A of GeDS, viewing ρ_i , from (1), as the observations y_i , $i = 1, \dots, N$.

Update the current set of knots as, $\mathbf{t}_{k+1,n+1} = \mathbf{t}_{k,n+1} \cup \delta^*$. Set, $\mathbf{t}_{k,n} \leftarrow \mathbf{t}_{k+1,n+1}$ and go back to Step 1.

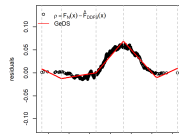




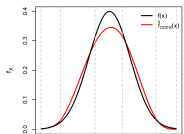
$$t_{k,n} = (a = -3.31, -2.37, 0.63, 2.24, b = 3.24)$$



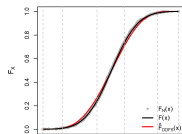
$$t_{k,n+1} = (a = -3.31, -2.37, 0.63, 2.24, b = 3.24)$$



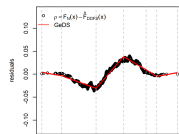
$$t_{1,k;2} = (a = -3.31, -2.37, -0.7, 0.63, 2.24, b = 3.24)$$



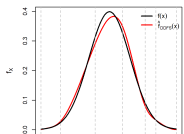
$$t_{k,n} = (a = -3.31, -2.37, -0.7, 0.63, 2.24, b = 3.24)$$



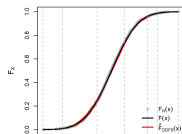
$$t_{k,n+1} = (a = -3.31, -2.37, -0.7, 0.63, 2.24, b = 3.24)$$



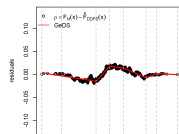
$$t_{1,k;2} = (a = -3.31, -2.37, -0.7, 0.63, 1.72, 2.24, b = 3)$$



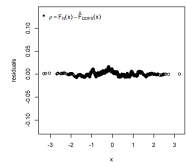
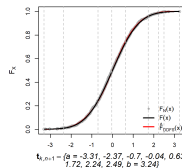
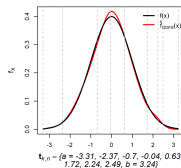
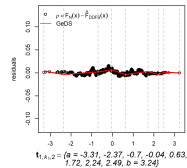
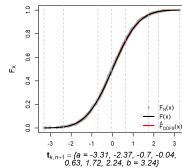
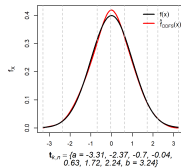
$$t_{k,n} = (a = -3.31, -2.37, -0.7, 0.63, 1.72, 2.24, b = 3.24)$$



$$t_{k,n+1} = (a = -3.31, -2.37, -0.7, 0.63, 1.72, 2.24, b = 3.24)$$



$$t_{1,k;2} = (a = -3.31, -2.37, -0.7, 0.63, 1.72, 2.24, b = 3.04)$$



Test case	Density	Test case	Density
Gaussian	$N(0, 1)$	Merton's jump diffusion	$\sigma = 0.08, \lambda = 3 \mu_J = -0.01,$ $\sigma_J = 0.4, \Delta_t = 1/4$
Student- t	$t_v, v = 6$	Kou's double exponential	$\sigma = 0.04, \lambda = 2 p_{up} = 0.4,$ $\eta_1 = 3, \eta_2 = 5, \Delta_t = 1/4$
Exponential	$\text{Exp}(\lambda), \lambda = 1, x \geq 0$	Generalized extreme value	$\frac{1}{\sigma} \exp\left(-\left(1 + k \frac{x-\mu}{\sigma}\right)^{-\frac{1}{k}}\right) (1 + k \frac{x-\mu}{\sigma})^{-1-\frac{1}{k}}$ $k = -0.5, 0 \text{ and } 0.5, \mu = 1, \sigma = 0, x > 0$
Chi-square	$\chi_k^2, k = 4, x > 0$	MixGauss	$0.15N(-0.25, 1/3) + 0.85N(3.25, 1)$
Gamma	$x \geq 0, k = 9, \theta = 0.5$	Mix1d	$0.8\chi^2(3) + 0.2N(7, 1)$
Weibull	$x \geq 0, \lambda = 1, k = 5$	MixGauss2	$\frac{5}{6}N(3, 1) + \frac{5}{36}N(8, (1/3)^2) + \frac{1}{36}N(10, (1/9)^2)$
Log-normal	$x > 0, \mu = 0, \sigma = 1$	Bimodal	$\frac{1}{2}N\left(0, \left(\frac{1}{10}\right)^2\right) + \frac{1}{2}N(5, 1)$
Nakagami	$\frac{2\mu^\mu x^{2\mu-1}}{\Gamma(\mu)\omega^\mu} \exp\left(-\frac{\mu}{\omega}x^2\right), x > 0, \mu = \omega = 2$	Separated bimodal	$\frac{1}{2}N(-2, \left(\frac{1}{2}\right)^2) + \frac{1}{2}N(2, \left(\frac{1}{2}\right)^2)$
Kurtotic unimodal	$\frac{2}{3}N(0, 1) + \frac{1}{3}N\left(0, \left(\frac{1}{10}\right)^2\right)$	Skewed bimodal	$\frac{3}{4}N(0, 1) + \frac{1}{4}N\left(\frac{3}{2}, \left(\frac{1}{3}\right)^2\right)$
Outlier	$\frac{1}{10}N(0, 1) + \frac{9}{10}N\left(0, \left(\frac{1}{10}\right)^2\right)$	Trimodal	$\frac{1}{3} \sum_{k=0}^2 N(80k, (k+1)^4)$
Skewed unimodal	$\frac{1}{5}N(0, 1) + \frac{1}{5}N\left(\frac{1}{2}, \left(\frac{2}{3}\right)^2\right) + \frac{3}{5}N\left(\frac{13}{12}, \left(\frac{5}{9}\right)^2\right)$	Smooth comb	$\sum_{k=0}^5 \frac{2^{5-k}}{63} N\left(\frac{65-96/2^k}{21}, \left(\frac{32/63}{2^k}\right)^2\right)$
Strongly skewed	$\sum_{k=0}^7 \frac{1}{8} N\left(3\left(\left(\frac{2}{3}\right)^k - 1\right), \left(\frac{2}{3}\right)^{2k}\right)$	Claw	$\frac{1}{2}N(0, 1) + \sum_{k=0}^4 \frac{1}{10} N\left(\frac{k}{2} - 1, \left(\frac{1}{10}\right)^2\right)$

We assess the goodness-of-fit over a regular grid of K evaluation points $x_1, \dots, x_K$ with uniform spacing $\Delta x = x_{i+1} - x_i$, based on:

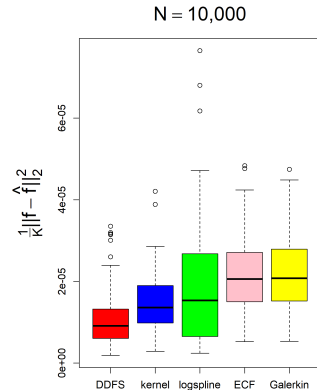
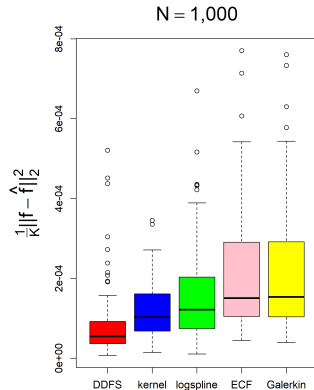
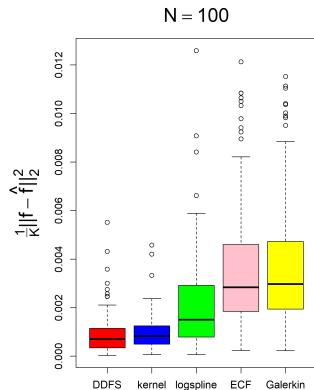
✱ Mean Integrated Squared Error (MISE)

$$\mathbb{E} \left[\int (\hat{f}(x; N) - f(x))^2 dx \right] \approx \Delta x \sum_{i=1}^K (\hat{f}(x_i; N) - f(x_i))^2$$

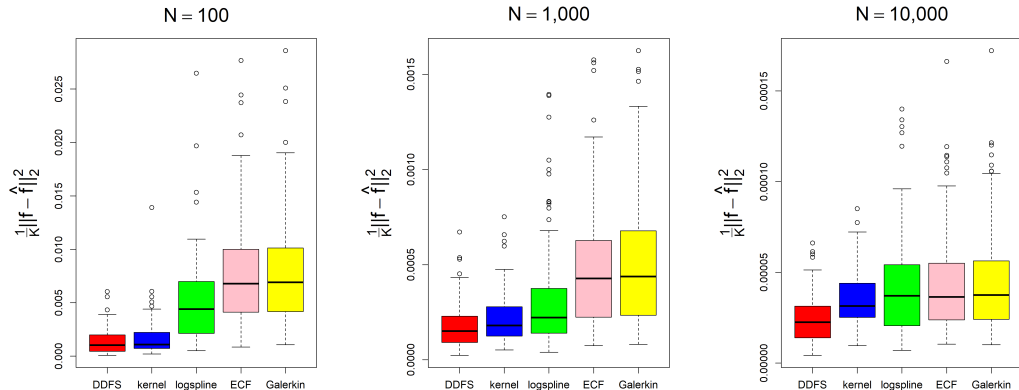
✱ Roughness, $R(f)$

$$R(f) = \int (f''(x))^2 dx \approx \Delta x \sum_{i=2}^{K-1} \left(\frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{(\Delta x)^2} \right)^2$$

MISE boxplots - Generalized Extreme Value, Type I (Gumbel)

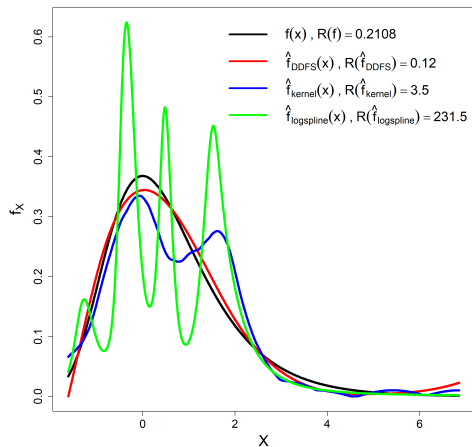


MISE boxplots - Generalized Extreme Value, Type III (Weibull)

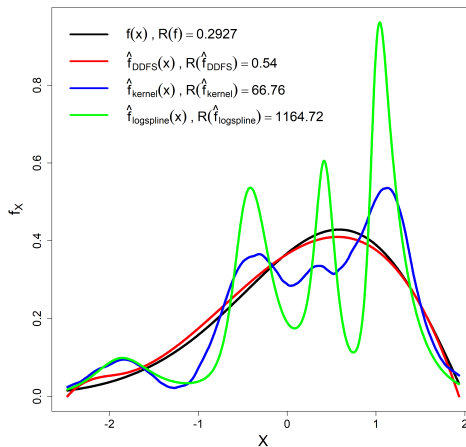


Roughness - Generalized Extreme Value, Type I and III (pdf)

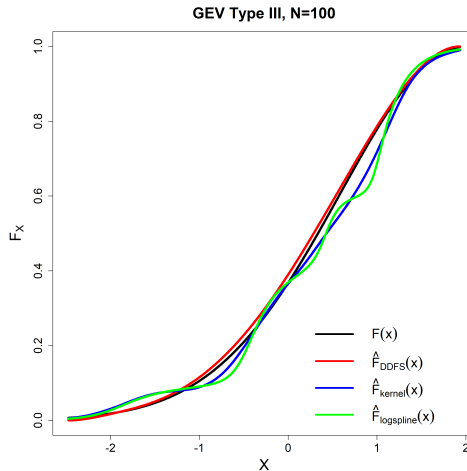
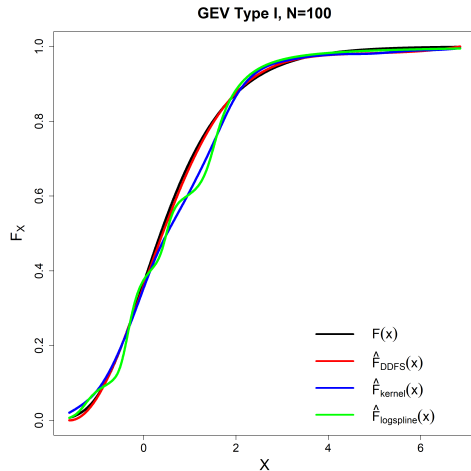
GEV Type I, N=100



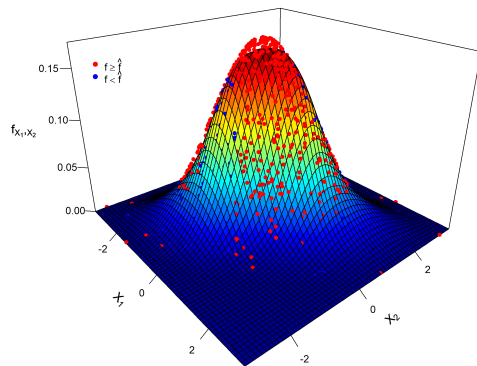
GEV Type III, N=100



Roughness - Generalized Extreme Value, Type I and III (cdf)

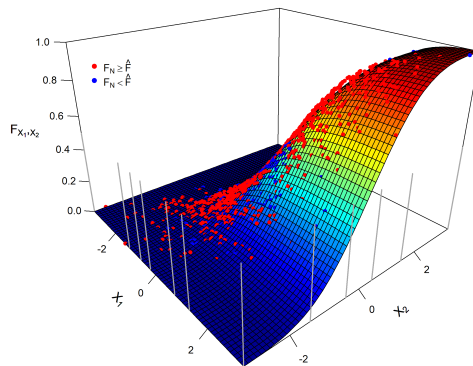


Bivariate Gaussian, $\rho = 0.5$

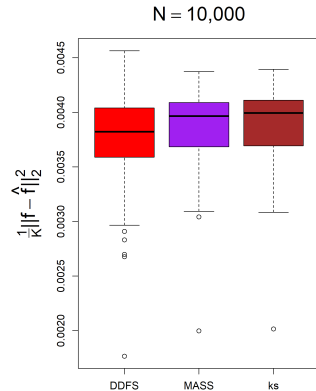
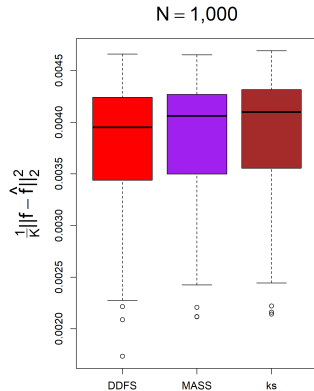
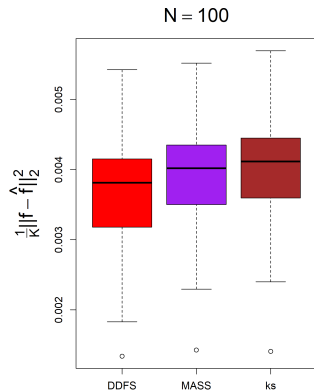


$$\mathbf{t}_{1; k_1, n_1+1} = \{a = -3.27, -1.24, -0.68, -0.23, 0.8, 1.37, b = 3.12\}$$

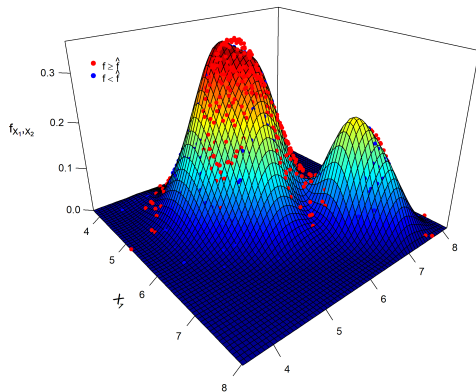
$$\mathbf{t}_{2; k_2, n_2+1} = \{a = -3.12, -1.46, -0.41, 0.43, 1.48, b = 3.45\}$$



MISE boxplots - Bivariate Gaussian, $\rho = 0.5$

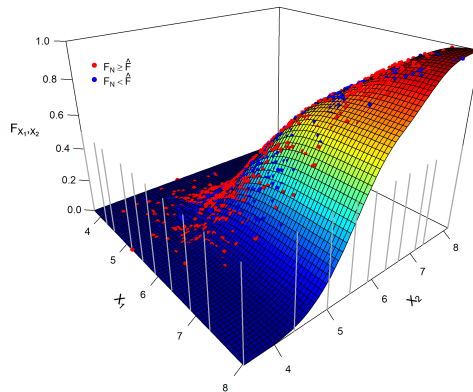


Bivariate Bimodal Mixed Gaussian

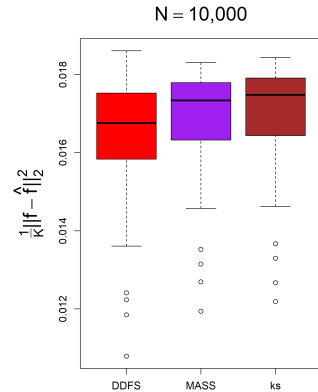
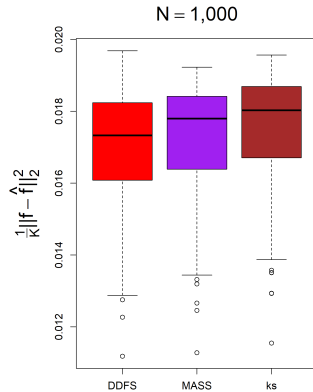
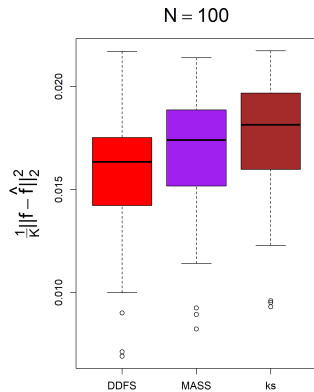


$$\mathbf{t}_{1; k_1, n_1+1} = \{a = 3.74, 4.26, 4.55, 5.18, 5.54, 5.92, 6.39, 6.87, 7.33, b = 8.02\}$$

$$\mathbf{t}_{2; k_2, n_2+1} = \{a = 3.48, 4.36, 5, 5.49, 5.9, 6.34, 6.68, 7.14, 7.38, b = 8.14\}$$



Bivariate Bimodal Mixed Gaussian



Loss data modelling

Proliferation of composite and mixture (parametric) models proposed for the modelling of insurance loss data (Marambakuyana and Shongwe, 2024).

✱ Estimation often requires **large samples to converge** (data availability might be a problem) + **high computational burden**:
—→ many actuarial studies only fit the model once (on the whole dataset) and perform backtesting on a static forecast (see, e.g., Abu Bakar et al., 2015);

✗ *rolling-window historical simulation backtesting.*

Some frequently considered loss-datasets are:

1. *Danish Fire Insurance.*
2. *Norwegian Fire Insurance Data.*
3. *US Allocated Loss Adjustment Expenses.*

➤ Assess model reliability via **rolling window back-testing** (Basel III, EIOPA):
Proportion of Failures (Kupiec, 1995), *Conditional Coverage* (Christoffersen, 1998),
Dynamic Quantile (Engle and and, 2004).

Danish Fire Insurance

VaR Backtesting: Actual Claims vs. VaR Forecast

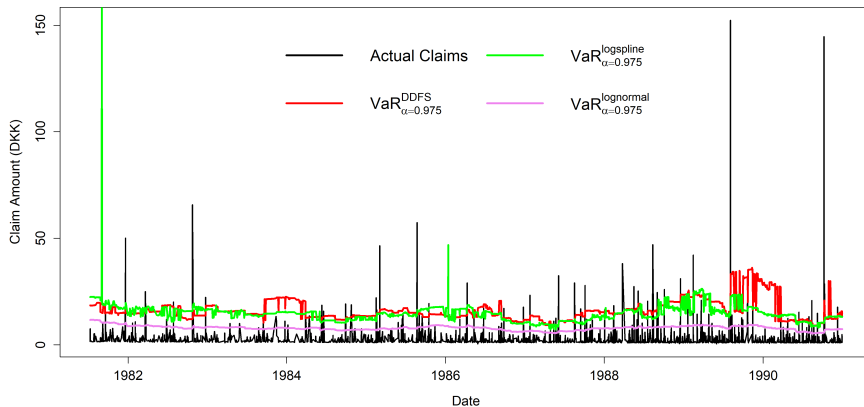


Table 1: Backtesting statistics for VaR models (data = danish).

	$\alpha = 0.975$				$\alpha = 0.99$			
	Viol. rate	UC	CC	DQ	Viol. rate	UC	CC	DQ
ddfs	0.0250	0.9946	0.1776	0.0244	0.0103	0.9024	0.7819	0.0349
kernel	0.0259	0.7931	0.0623	0.0001	0.0116	0.4586	0.5600	0.0006
logspline	0.0245	0.8867	0.4458	0.1013	0.0120	0.3462	0.4617	0.0256
lognormal	0.0531	0.0000	0.0000	0.0000	0.0397	0.0000	0.0000	0.0000
gamma	0.0437	0.0000	0.0000	0.0000	0.0361	0.0000	0.0000	0.0000

Norwegian Fire Insurance Data

VaR Backtesting: Actual Claims vs. VaR Forecast

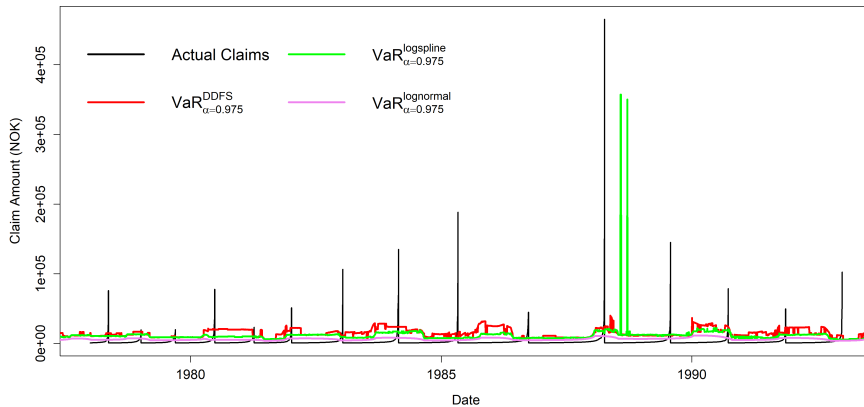


Table 2: Backtesting statistics for VaR models (data = norwegian).

	$\alpha = 0.975$				$\alpha = 0.99$			
	Viol. rate	UC	CC	DQ	Viol. rate	UC	CC	DQ
ddfs	0.0265	0.3816	0.0000	0.0206	0.0100	0.9832	0.0000	0.0001
kernel	0.0351	0.0000	0.0000	0.0078	0.0156	0.0000	0.0000	0.0000
logspline	0.0356	0.0000	0.0000	0.0074	0.0147	0.0001	0.0000	0.0000
lognormal	0.0583	0.0000	0.0000	0.0032	0.0390	0.0000	0.0000	0.0000
gamma	0.0466	0.0000	0.0000	0.0048	0.0336	0.0000	0.0000	0.0000

US Allocated Loss Adjustment Expenses

VaR Backtesting: Actual Claims vs. VaR Forecast

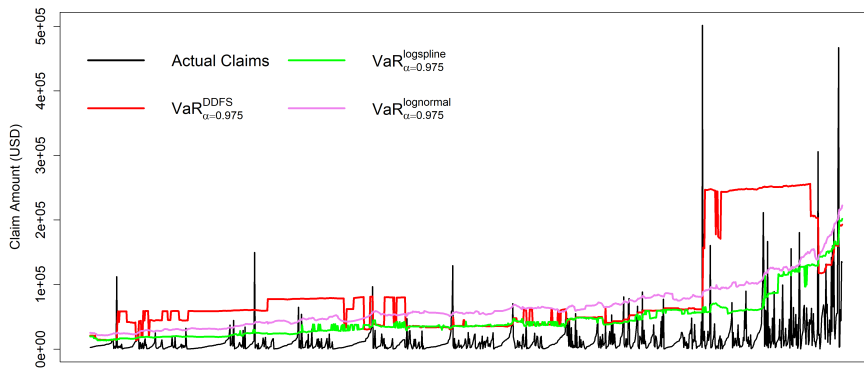


Table 3: Backtesting statistics for VaR models (data = lossalae).

	$\alpha = 0.975$				$\alpha = 0.99$			
	Viol. rate	UC	CC	DQ	Viol. rate	UC	CC	DQ
ddfs	0.0256	0.8923	0.0000	0.9663	0.0112	0.6757	0.0000	0.6161
kernel	0.0448	0.0001	0.0000	0.4172	0.0224	0.0002	0.0000	0.2674
logspline	0.0448	0.0001	0.0000	0.4292	0.0272	0.0000	0.0000	0.0133
lognormal	0.0200	0.2410	0.0218	0.9555	0.0072	0.2950	0.5413	0.6525
gamma	0.0488	0.0000	0.0000	0.4018	0.0360	0.0000	0.0000	0.0040

-  Abu Bakar, S., Hamzah, N., Maghsoudi, M., & Nadarajah, S. (2015). Modeling loss data using composite models. *Insurance: Mathematics and Economics*, 61, 146–154. <https://doi.org/https://doi.org/10.1016/j.insmatheco.2014.08.008>
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